

Shaqayeq Matlabi

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Summary

Machine Learning Engineer with 2 years experience and with expertise in building and deploying deep learning models using PyTorch and TensorFlow. Experienced in data preprocessing, model optimization, and designing RESTful APIs. Proven track record in medical image segmentation, language modeling, and speech-to-text applications.

Work Experience

Startup (Confidential), AI Engineer

Remote - Part time
July 2025 Present

- Developed a Retrieval-Augmented Generation (RAG) system integrating vector search with large language models (LLMs) to deliver domain-specific answers in real time.
- Built and optimized an infant cry analysis model using deep audio feature extraction to classify emotional states such as hunger, pain, and sleepiness.
- Scraped and curated datasets from 10 different websites, performing data cleaning, labeling, and preprocessing for training and evaluation.
- Deployed and monitored both systems through API endpoints, enabling scalable real-world inference and performance tracking.

Product Manager Accelerator, AI Engineer Intern

United States
June 2025 Aug 2025

- Designed and implemented AI-powered tools to enhance product management workflows using RAG and Transformer-based models.
- Built intelligent agents that interact with unstructured documents and respond to user queries in real-time across platforms.
- Collaborated remotely with cross-functional teams to integrate LLMs via APIs and monitor usage with Grafana dashboards.

British Airways, Data Science Intern

United Kingdom
Aug 2024

- Completed tasks on data wrangling, predictive analytics, and visualization, demonstrating proficiency in handling data-driven projects.
- Analyzed and modeled real-world airline data to enhance user experience and operational efficiency.

Education

MSc Payame-Noor University, **Applied Mathematics** (Optimization)

BMath Teacher Training University, **Mathematics Teaching**

Projects

Discord RAG Agent

[GitHub](#)

- Developed an intelligent **RAG-based assistant** for Discord that answers user questions using PDF/text documents.
- Integrated **MiniLM + FAISS** for semantic search and **OpenRouter GPT** for natural response generation.
- Enabled production-ready monitoring with **Prometheus and Grafana** dashboards for real-time performance tracking.

Handwritten Persian/Arabic Math Expressions Dataset (HME)

[Kaggle Dataset](#)

- Curated 26,141 annotated samples of handwritten math expressions with Persian/Arabic numerals

- Manually labeled all samples with LaTeX codes, including 6,960 complex formulas (integrals, matrices)
- Created comprehensive benchmark dataset for multilingual OCR research in mathematical notation

Handwritten Mathematical Expression Recognition with Persian Numerals

(Research Paper - Under Review)

[Preprint](#) | [GitHub](#)

- Proposed novel **ResNet-18/Transformer** hybrid architecture for HME recognition
- Demonstrated cross-cultural OCR applicability through comprehensive experiments
- Published dataset and code openly to support **multilingual AI research**

Medical Image Segmentation

[GitHub](#) | [Kaggle Notebook](#)

- Built a multi-class **segmentation** model in **PyTorch** for MRI scans with 85 percent mIoU accuracy.

Speech-to-Text

[GitHub](#)
[HuggingFace space](#)

- Designed and **deployed** a speech-to-text web app using **Transformers**, achieving a WER of 0.66.

Language Modeling

[GitHub](#)
[HuggingFace space](#)

- Crafted a sophisticated **machine learning model** utilizing **PyTorch**; trained on the **WikiText-2** dataset to improve text generation accuracy within less than 5 percent error margin compared to industry standards.

Image Captioning

[GitHub](#)
[HuggingFace space](#)

- Engineered an innovative web application utilizing a **deep learning model** via **PyTorch**, generating accurate image captions with a performance score of 2.8 on the **Flickr dataset** to enhance user engagement.

Skills and Technologies

Languages: Python, MATLAB, SQL

Machine Learning: Supervised and Unsupervised Learning, Deep Learning, NLP, Computer Vision, RAG, Langchain

Tools and Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras

Cloud and Deployment: AWS, Docker, Git, Hugging Face Spaces, continuous integration and delivery

Data Visualization: Tableau, Power BI, Grafana

Languages

Persian: C1

Turkish: B2

English: B2